

Description

The pediatrician suspects measles and orders a Real-time RT-PCR to confirm, and provides instruction for home isolation to prevent potential spread.

A nurse weighs the patient, measures his length, and charts the data in the patient's record at 8:35. This information may be used by the physician if ordering the take-home prescription. At 8:55 AM, the physician assigns the patient the working diagnoses of measles: (ICD-10 CM diagnosis code of B05.9 Measles without complication), Conjunctivitis (ICD-10 CM diagnosis code of H10.10 Acute atopic conjunctivitis, unspecified eye), Fever (ICD-10 CM diagnosis code of R50.9 Fever, unspecified), and Cough (ICD-10 CM diagnosis code of R05.9 Cough, unspecified) are documented as final in the patient's electronic medical record. Suspecting Measles, the provider takes a nasal swab to send for Real-time RT-PCR (CPT-4 code of 87798 Infectious agent detection by nucleic acid). The provider recommends isolation at home, providing education to the parents (CPT-4 code of 98960 Education and Training for Patient Self-Management). The visit is closed at 9:05 am after the nurse collects the sample to send to the lab.

NIST Primary Care is an outpatient facility that routinely sends electronic syndromic surveillance data to the City Health Department (NIST PH SS FAC 1) in accordance with a city regulation. At 6:00 PM, the facility's electronic health record module for syndromic surveillance data assembles and transmits a Discharge ADT message with height and weight information, as well as the working and final diagnoses for this patient encounter. The patient name and birthdate are de-identified in the submission, so no date of birth is included, and the name is represented with name type 'S' to indicate that the name is empty for anonymity.

Comments

The dates and times in this test case illustrate the sequence of clinical and messaging events. Since the exact dates and times are not reproducible when modeling the test case with EHR technology, only date and time format will be validated within tester submitted test data.

The dates within the message and across the messages in this test case will be subject to visual inspection to verify data consistencies and the ability to conform to the requirement to send this message within 12 hours of the event triggering this message.

Pre-condition

A04-Registration message is sent before the A03-Discharge message.

Post-Condition

An A03-Discharge message has been triggered.

Test Objectives

This test step examines a Health IT Module's ability to create an ADT^A03 Disposition message within HL7 Version 2.5.1 Implementation Guide: Syndromic Surveillance, Release 1 - US Realm Standard for Trial Use July 2019 conformance requirements.

- Setting is Ambulatory.
- Chief complaint is captured as narrative in PV2-3 Admit Reason
- **Patient name is de-identified in the submissions:** Per ASTP/ONC, for certification testing HIT developers must demonstrate that their system supports the Patient Name and de-identified data in PID-5 (this policy is conformant to the named standard HL7 Version 2.5.1 Implementation Guide: Syndromic Surveillance, Release 1 - US Realm Standard for Trial Use July 2019). The patient name presented in a de-identified manner is required for this Test Step.
- **Diagnoses are submitted test using either ICD-10-CM or SNOMED-CT:** Per ASTP/ONC, for certification testing a Health IT Module can use either the ICD-10-CM or SNOMED CT® value sets in the submitted messages for all of the test steps in the Syndromic Surveillance Test Suite. Where the tool does not have test data that supports the Health IT Module's value set (either ICD-10-CM or SNOMED CT®), the developer of the Health IT Module must provide the codes and testers must perform a visual inspection of the messages for these test steps to ensure that equivalent and valid ICD-10-CM or SNOMED CT® are used to populate the messages. This policy is conformant to the named standard HL7 Version 2.5.1 Implementation Guide: Syndromic Surveillance, Release 1 - US Realm Standard for Trial Use July 2019). ICD-10-CM codes are provided/required in the Test Data for coding the diagnoses in this Test Case.
- Pediatric case where age is measured in months.
- Add Diagnoses
- Multiple Races
- Insurance unknown

Evaluation Criteria

Message Validation Report

Diagnosis code (DG1-3) validation is validated as follows:

The table below summarizes the format representations that are accepted for ONC Health IT Module Certification. Rows 1 and 2 show the formats as required by the implementation guide and standard terminology rules and best practices. The last 3 columns show acceptable alternative formats. For this test case one of the following values shall be present in the message:

DG1(1)-3

Code	Concept	Code System	DG1-3 Message Snippet	NIST
Required Formats based on Recognized Terminology Standards Rules				
B059	Measles without complication	I10C	B059^Measles without complication^I10C	Yes
14189004	Measles (disorder)	SCT	14189004^Measles (disorder)^SCT	Yes
Additional Alternate Allowable Formats and codes based on similar concepts, Common Industry Practice and CDC SMEs Preferences				
B05.9	Measles without complication	I10C	B05.9^Measles without complication^I10C	Yes
B059	Measles without complication	I10	B059^Measles without complication^I10	No
B05.9	Measles without complication	I10	B05.9^Measles without complication^I10	No
74325003	Disease caused by Morbillivirus (disorder)	SCT	74325003^Disease caused by Morbillivirus (disorder)^SCT	Yes

DG1(2)-3

Code	Concept	Code System	DG1-3 Message Snippet	NIST
Required Formats based on Recognized Terminology Standards Rules				
H1010	Acute atopic conjunctivitis, unspecified eye	I10C	H1010^Acute atopic conjunctivitis, unspecified eye^I10C	Yes

53726008	Acute conjunctivitis (disorder)	SCT	14189 53726008 004^Acute conjunctivitis (disorder)^SCT	Yes
Additional Alternate Allowable Formats and codes based on similar concepts, Common Industry Practice and CDC SMEs Preferences				
H10.10	Acute atopic conjunctivitis, unspecified eye	I10C	H10.10^Acute atopic conjunctivitis, unspecified eye^I10C	Yes
H1010	Acute atopic conjunctivitis, unspecified eye	I10	H1010^Acute atopic conjunctivitis, unspecified eye^I10	No
H10.10	Acute atopic conjunctivitis, unspecified eye	I10	H10.10^Acute atopic conjunctivitis, unspecified eye^I10	No
9826008	Conjunctivitis (disorder)	SCT	9826008^Conjunctivitis (disorder)^SCT	Yes
DG1(3)-3				
Code	Concept	Code System	DG1-3 Message Snippet	NIST
Required Formats based on Recognized Terminology Standards Rules				
R509	Fever, unspecified	I10C	R509^Fever, unspecified^I10C	Yes
416113008	Febrile disorder (disorder)	SCT	416113008^Febrile disorder (disorder)^SCT	Yes
Additional Alternate Allowable Formats and codes based on similar concepts, Common Industry Practice and CDC SMEs Preferences				
R50.9	Fever, unspecified	I10C	R50.9^Fever, unspecified^I10C	Yes
R509	Fever, unspecified	I10	R509^Fever, unspecified^I10	No
R50.9	Fever, unspecified	I10	R50.9^Fever, unspecified^I10	No
DG1(4)-3				
Code	Concept	Code System	DG1-3 Message Snippet	NIST
Required Formats based on Recognized Terminology Standards Rules				
R059	Cough, unspecified	I10C	R059^Cough, unspecified^I10C	Yes
417850002	Respiratory tract congestion and cough (disorder)	SCT	417850002^Respiratory tract congestion and cough (disorder)^SCT	Yes
Additional Alternate Allowable Formats and codes based on similar concepts, Common Industry Practice and CDC SMEs Preferences				
R05.9	Cough, unspecified	I10C	R05.9^Cough, unspecified^I10C	Yes
R059	Cough, unspecified	I10	R059^Cough, unspecified^I10	No
R05.9	Cough, unspecified	I10	R05.9^Cough, unspecified^I10	No

The NIST column indicates what the NIST Tool Validation will accept in context-based testing (used for ONC Certification). A 'Yes' in the NIST column signifies that the tool will accept the format and not throw an error. A 'No' in the NIST column signifies that the tool will throw an error (specifically for DG1-3.3). However, based on the recommendation from the CDC SMEs and approval from ASTP/ONC the format is acceptable. The NIST test cases documents the instances where the test proctor can grant an exception when the tool issues and error.

Notes

Codes are provided in the test data for coding the diagnoses and procedures. The Tool is designed to accept those codes without generating an error. If the HIT developer uses a different but equivalent coded value within the same code system than the ones provided/accepted, the Tester may ignore errors generated by the Test Tool related to incorrect code/name of coding system when the coded data used in the message are determined via visual inspection to be valid data.

Messages in this test case will be used in a Manual Test Step under functional requirement testing.